

# CS 320: Lab04: Web Applications II Lab Sign-Off

## Tests for Web Applications II Lab Sign-Off

- 1) Pull up **Lab04** in IntelliJ
- 2) Pull up **Index page**: Verify that there are links/buttons for **AddNumbers**, **MultiplyNumbers**, and **GuessingGame**.
- 3) Go to **Add Numbers** from **Index page**: **AddNumbers** page appears, with three number fields.
- 4) Enter **three numbers**: Get correct sum.
- 5) Delete **third number**: Get correct error message: **"Please enter 3 numbers"** (required), fields have original values (extra credit).
- 6) Replace **third number** with **"abc"**: Get correct error message: **"Invalid double"** (required), fields have original values (extra credit).
- 7) Go back to the **Index Page**.
- 8) Go to **Multiply Numbers**: **MultiplyNumbers** page appears, with two number fields.
- 9) Enter **two numbers**: Get correct product.
- 10) Delete **first number**: Get correct error message: **"Please enter 2 numbers"** (required), all fields have original values (extra credit).
- 11) Replace **first number** with **"abc"**: Get correct error message: **"Invalid double"** (required), all fields have original values (extra credit).
- 12) Go back to **Index Page**.
- 13) Go to **Guessing Game**: **GuessingGame** page appears, **"Start Game"** displayed.
- 14) Use **33** as the number: Press correct buttons, application eventually guesses 33. Sequence of guesses: **50 -> 25 -> 37 -> 31 -> 34 -> 32 -> 33**.
- 15) Pull up **Numbers.java**: Verify encapsulation. Model attributes are private, and public constructor(s), getters, setters have been provided.
- 16) Pull up **addNumbersServlet.java**: Verify **ONLY error message** and **model** parameters are being set in the HTTP request.
- 17) In **addNumbersServlet.java**: Verify MVC is followed. **NO** direct calls to **Numbers** model setter methods. Only controller methods used to change **Numbers** attributes.
- 18) In **multiplyNumbersServlet.java**: Verify MVC is followed. **NO** direct calls to **Numbers** model setter methods. Only controller methods used to change **Numbers** attributes.
- 19) Pull up **addNumbers.JSP**: Verify it **ONLY** pulls model data directly from **Numbers.java** model.

20) Run JUnit tests for **NumbersController.java** and **Numbers.java** methods: Test cases exist and pass. Complete test cases cover **constructor(s)**, **getters**, **setter(s)**, and **add** and **multiply** methods.

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